Question 055 | Question bank page 6 | Q.1

Develop an ASP.NET Core Web API to retrieve and return student details in JSON format.

Project type: ASP.NET Core Web API

Steps to perform

1. In Visual Studio create an ASP.NET Core Web API project.

2. Add Models and Controllers files; configure required services in Program.cs.

3. If EF Core is used, install matching provider packages, set the connection string and apply migrations.

4. Build and run the application.

5. Test GET/POST/PUT/DELETE endpoints using Postman or Swagger where configured.

Closest supplied practical:] } Create a simple Web API that returns Student data in JSON format.

Source: Dot_Net_Practicals_6_to_10_Organized_v2.pdf

Provided creation instructions and source files

```
Creation steps: Create the StudentAPI project. Add the Student model, API controller and launch
settings;
run the app and request the Student endpoint.
Filename: Student.cs    Location: Models/Student.cs
namespace StudentAPI.Models
{
    public class Student
    {
        public int Id { get; set; }
        public string Name { get; set; }
        public string Course { get; set; }
        public int Age { get; set; }
    }
}
Filename: StudentController.cs    Location: Controllers/StudentController.cs
using Microsoft.AspNetCore.Http;
using Microsoft.AspNetCore.Mvc;
using StudentAPI.Models;
namespace StudentAPI.Controllers
{
    [Route("api/[controller]")]
    [ApiController]
    public class StudentController : ControllerBase
    {
        [HttpGet]
        public List<Student> GetStudents()
        {
            return new List<Student>()
            {
                new Student
                {
                    Id = 1,
                    Name = "Akash",
                    Course = "BScIT",
                    Age = 20
                },
                new Student
                {
                    Id = 2,
                    Name = "Aditya",
                    Course = "BScCS",
                    Age = 19
                }
            };
        }
    }
}
Filename: LaunchSettings.json    Location: Properties/launchSettings.json
{
  "$schema": "https://schemastore.org",
  "profiles": {
    "http": {
      "commandName": "Project",
      "dotnetRunMessages": true,
      "launchBrowser": true,
      "applicationUrl": "http://localhost:5000",
```

```
      "environmentVariables": {
        "ASPNETCORE_ENVIRONMENT": "Development"
      }
    },
    "https": {
      "commandName": "Project",
      "dotnetRunMessages": true,
      "launchBrowser": true,
      "applicationUrl": "https://localhost:7124;http://localhost:5000",
      "environmentVariables": {
        "ASPNETCORE_ENVIRONMENT": "Development"
      }
    }
  }
}
```

Expected result / verification

The application should meet the exact question requirements. Record actual output or screenshots after running; no execution has been claimed in this manual.